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Date: July 27, 2026

Editor-in-Chief

Machine Learning with Applications
Elsevier

Dear Editor-in-Chief,

Please find enclosed our manuscript entitled “**Counterfactual Evaluation of Causal Faithfulness in GINE Explanations for Molecular Toxicity Prediction**” for consideration as a Research Paper in the Explainability/Interpretable Machine Learning section of *Machine Learning with Applications*.

In this work, we investigate whether standard Graph Neural Network (GNN) training induces causally faithful explanations, evaluated through a counterfactual-sufficiency protocol on molecular toxicity benchmarks. Our work addresses the critical gap between plausible (correlational) and causally faithful explanations in graph neural networks, which is essential for trustworthy AI in safety-critical domains such as AI-aided drug discovery.

Our empirical findings provide a rigorous, replicated negative result that is highly relevant to the machine learning community: we demonstrate that standard empirical risk minimization on a state-of-the-art Graph Isomorphism Network with Edge features (GINE) does not induce causal alignment between attributed toxicophores and model decision boundaries. This negative result is validated across five random seeds, three standard molecular datasets (BBBP, BACE, TOX21), and multiple evaluation metrics, providing robust evidence that challenges conventional explainability assumptions and motivates the development of more principled objectives for causal alignment in GNNs.

The manuscript fits the scope of *Machine Learning with Applications* as it:

1. Applies advanced machine learning methodologies to a high-impact real-world domain (molecular toxicity prediction).
2. Employs a rigorous empirical evaluation protocol with multi-seed statistical validation.
3. Addresses fundamental reliability challenges in explainable AI (XAI).
4. Provides actionable, evidence-based guidance for ML practitioners building safety-critical predictive systems.

Declaration of Competing Interest:

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability:

All benchmark datasets (BBBP, BACE, TOX21) are publicly available from MoleculeNet. Our complete codebase, multi-seed training scripts, verification harnesses, and raw experimental log JSONs are publicly available under the MIT License at:

<https://github.com/GauravPatil2515/Faithful-LLM-Augmented-Explainability-for-Graph-Neural-Nets>

We confirm that this work is original, has not been published elsewhere, and is not currently under consideration by any other journal.

Thank you very much for your time and consideration of our submission.

Sincerely,

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